

6. Best AC Null Detector

(a) Wheatstone Bridge: **Galvanometer or sensitive voltmeter**

(b) Hay Bridge: **Headphones or tuned amplifier**

(c) Schering Bridge: **Electrometer or electronic voltmeter**

(d) Maxwell Bridge: **Headphones or electronic voltmeter**

7. What type of bridge is best suited for measuring capacitance? **Schering Bridge**

8. What type of bridge is best suited for measuring the inductance of high-Q coils? **Hay Bridge**

9. What type of bridge, is best suited for measuring the inductance of low-Q coils? **Maxwell Bridge**

10. What type of bridge is best suited for measuring the power factor of a capacitor? **Schering Bridge**

11. A zero-center galvanometer with a range of 1-0-1 mA is used as the null detector in a de Wheatstone bridge. What type of circuit or device may be used to increase the sensitivity of the galvanometer so that 10 μ A will deflect the meter full scale? **Use a differential amplifier**

12. What type of device or circuit may be used with an ac Wheatstone bridge to drive a ground-referenced null indicator? **Use an amplifier or an isolation transformer to drive a ground-referenced null indicator**

Problems

$$1 - E_0 = E_3 - E_4$$

$$= E \left(\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right)$$

$$= 1,5 \left(\frac{16}{16 + 1,2} - \frac{5}{16 + 5} \right) = 0$$

$\therefore E_0 = 0$ $\therefore$ bridge is balanced

$$2 - E_0 = 1,5 \left(\frac{16}{16 + 1,2} - \frac{5,2}{5,2 + 10} \right)$$

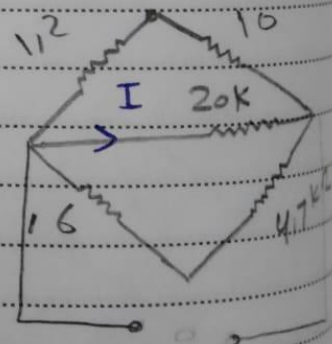
$$= -10,13 \text{ V}$$

$$3 - I = \frac{E_0}{20 \text{ k}\Omega}$$

$$E_0 = 1,5 \left(\frac{16}{16 + 1,2} - \frac{4,7}{10 + 4,7} \right)$$

$$= 102 \text{ V}$$

$$\therefore I = \frac{102}{20 \times 10^3} = 1 \text{ mA}$$

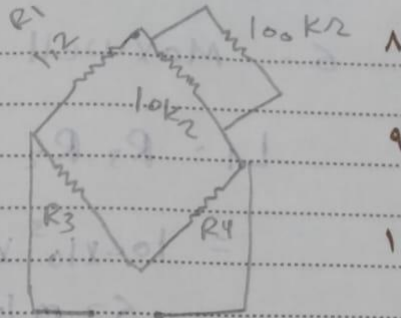


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$$4 - R_2 = \frac{10 \times 100}{10 + 100} = 9.09 \text{ K}\Omega$$

$$E_o = 1.5 \left[\frac{16}{1.2 + 16} - \frac{5}{5 + 9.09} \right]$$

$$= -1.03 \text{ V}$$



5- at null conduction:-

$$\frac{R_3}{R_1 + R_3} = \frac{R_4}{R_2 + R_4}$$

$$R_3 R_2 + R_3 R_4 = R_1 R_4 + R_3 R_4$$

$$\therefore R_1 R_4 = R_3 R_2$$

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$$\therefore R_x = \frac{R_2}{R_1} R_3$$

$$= \frac{10}{1.2} \times 1576 = 4.8 \text{ K}\Omega$$

6- Maxwell's bridge :-

$$L_1 = R_3 R_2 C_1$$

$$= 100 \times 10^3 \times 10 \times 10^3 \times 62 \times 10^{-12}$$
$$= 62 \text{ mH}$$

$$7- R_4 = \frac{R_2}{R_1} \times R_3$$

$$= \frac{10}{100} \times R_3 = 100 \text{ k}\Omega$$

$$8- \phi_1 = \omega R_1 C_1$$

$$= 2\pi f R_1 C_1$$

$$= 2 \times 3.14 \times 10^3 \times 10 \times 10^3 \times 62 \times 10^{-12}$$
$$= 3.9 \times 10^{-3}$$

9-

$$L_1 = \frac{R_2 R_3 C_1}{1 + \left(\frac{1}{Q}\right)^2}$$

$$Q = \frac{1}{\omega R_1 C_1}$$

$$= \frac{1}{2 \times 3.14 \times 10^3 \times 10^{-4} \times 93 \times 10^{-12}}$$

$$= 171.22$$

$$L_1 = \frac{10 \times 10^3 \times 100 \times 10^3 \times 92 \times 10^{-12}}{1 + \left(\frac{1}{171.22}\right)^2}$$

$$= 92 \text{ mH}$$

